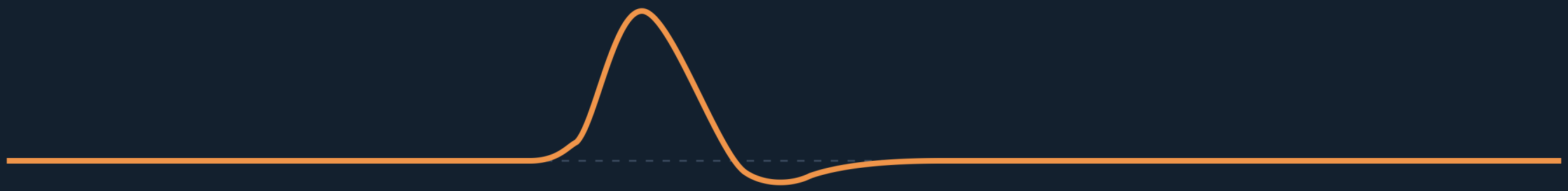
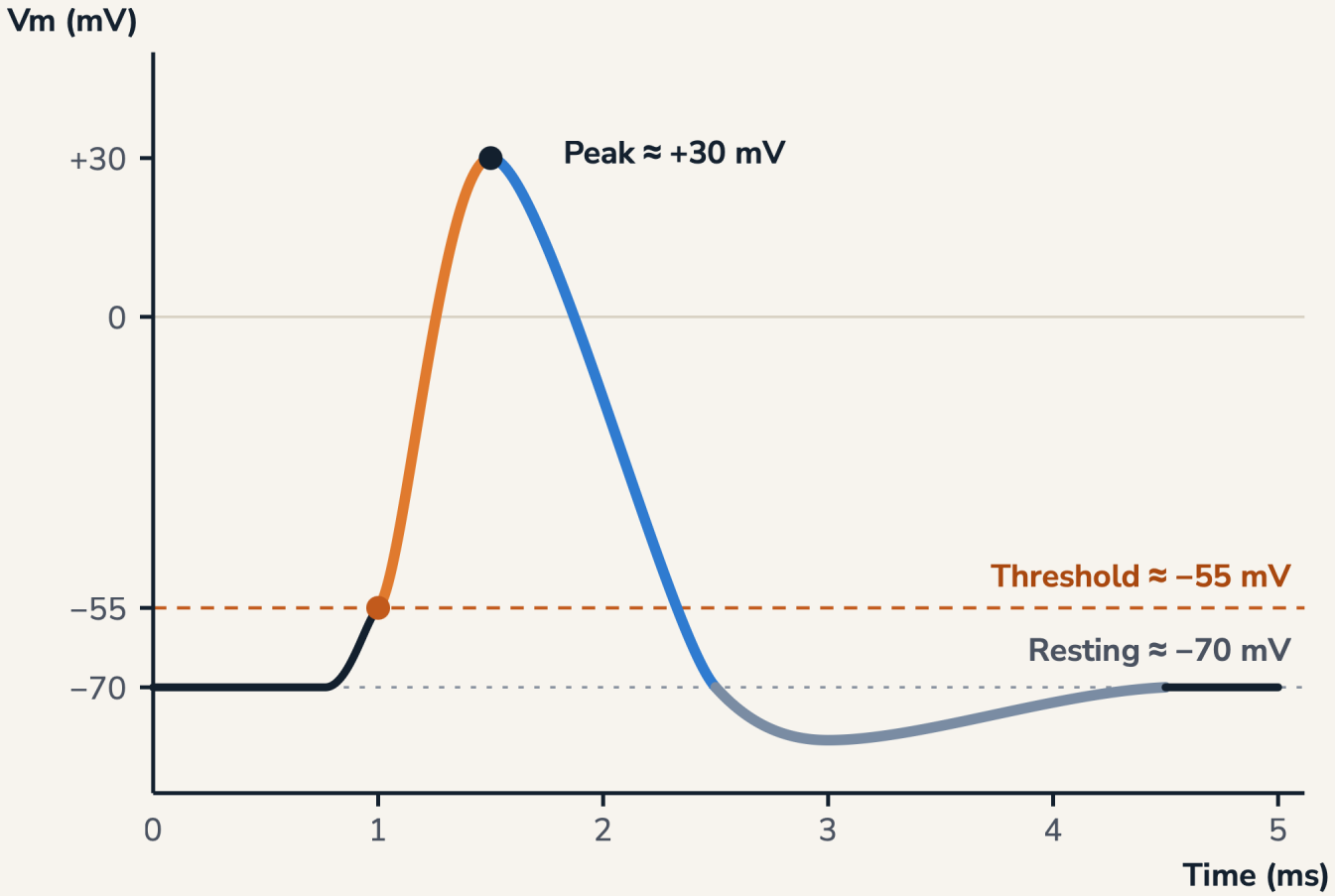
NEUROPHYSIOLOGY

The Action Potential

How voltage-gated Na^+ and K^+ channels turn a small nudge into an all-or-none electrical signal



Overview: reading the graph



Depolarization

Na^+ channels open and Na^+ rushes in. Vm swings from −55 to about +30 mV in under 1 ms.

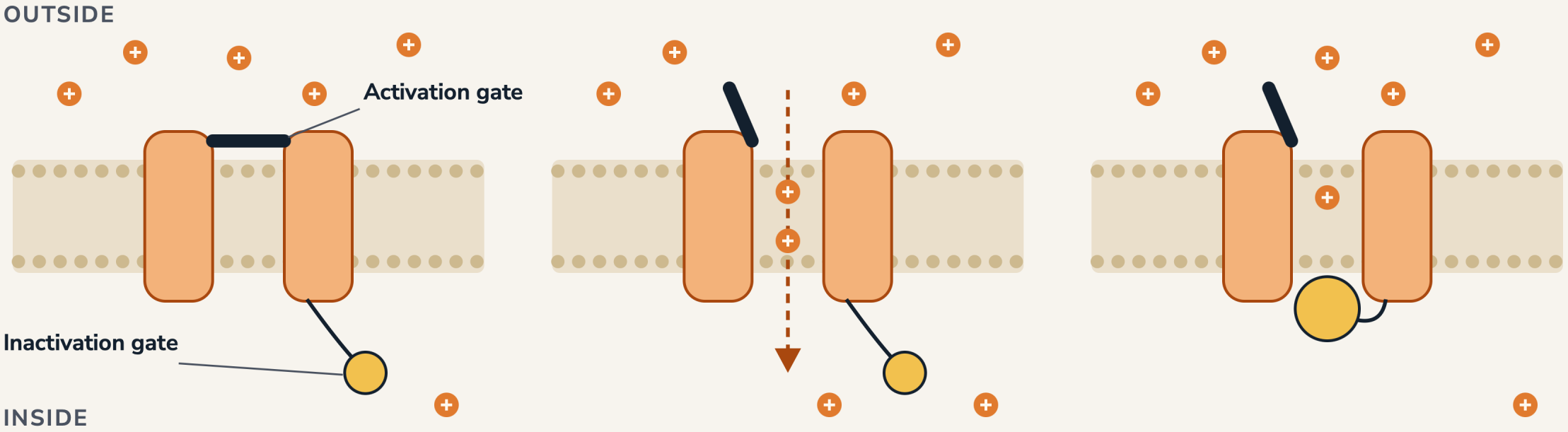
Repolarization

Na^+ channels inactivate while K^+ channels open. K^+ flows out and Vm falls back toward rest.

Hyperpolarization

K^+ channels close slowly, so Vm undershoots to about −80 mV before settling at rest.

Voltage-gated Na^+ channel: two gates



1 · RESTING

Closed

Activation gate shut, inactivation gate open. No Na^+ flows, but the channel is ready to fire.

2 · ACTIVATED

Open

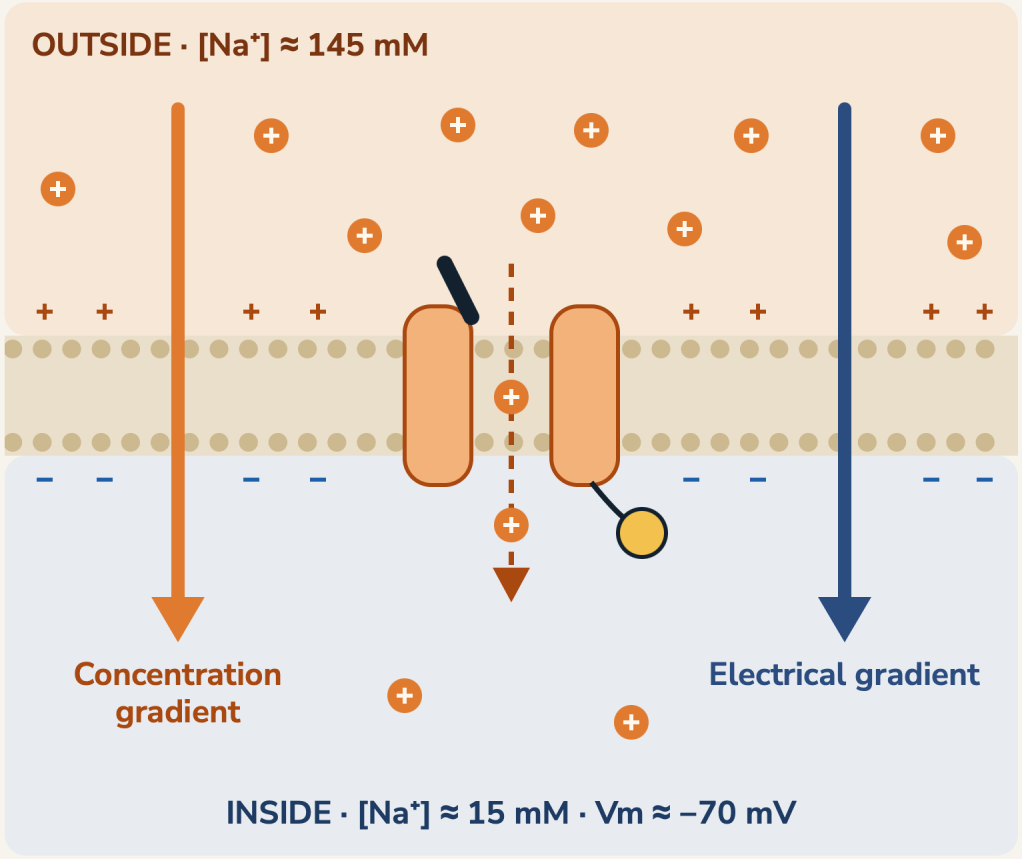
Reaching threshold flips the activation gate open within a fraction of a millisecond. Na^+ floods in.

3 · INACTIVATED

Plugged

About 1 ms later the inactivation “ball” blocks the pore. It resets only after the membrane repolarizes.

Na^+ channel function: a strong inward pull



DRIVING FORCE ON NA⁺

−130 mV

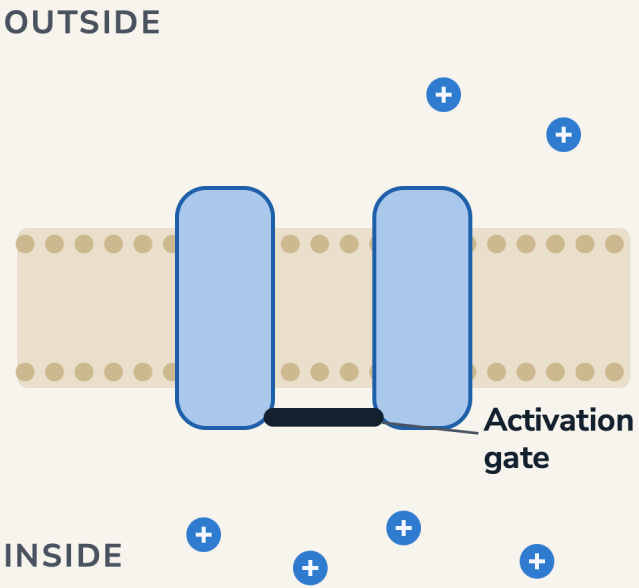
Vm − ENa = (−70) − (+60). Both gradients point inward, so the instant the channels open, Na^+ pours in.

POSITIVE FEEDBACK

Depolarize → Channels open → Na^+ enters

Each Na^+ that enters depolarizes the membrane further and opens more channels. The loop stops when channels inactivate and Vm nears ENa (+60 mV), where the driving force shrinks.

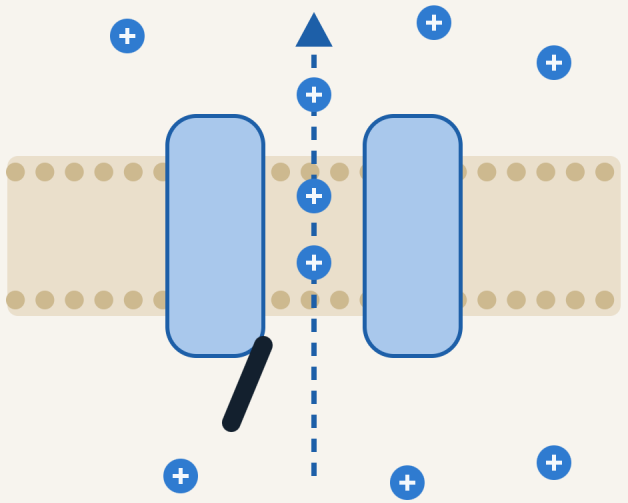
Voltage-gated K^+ channel: one slow gate



1 · AT REST

Closed

The single activation gate sits across the inner mouth of the pore.



2 · NEAR THE PEAK

Open

The gate swings open slowly. K^+ leaves the cell, pulling Vm back down.

One gate, no inactivation

Only an activation (n) gate. Once open, it stays open as long as the membrane is depolarized.

Delayed opening

Triggered by the same depolarization as Na^+ channels, but much slower, so K^+ current builds as the spike peaks.

Outward driving force

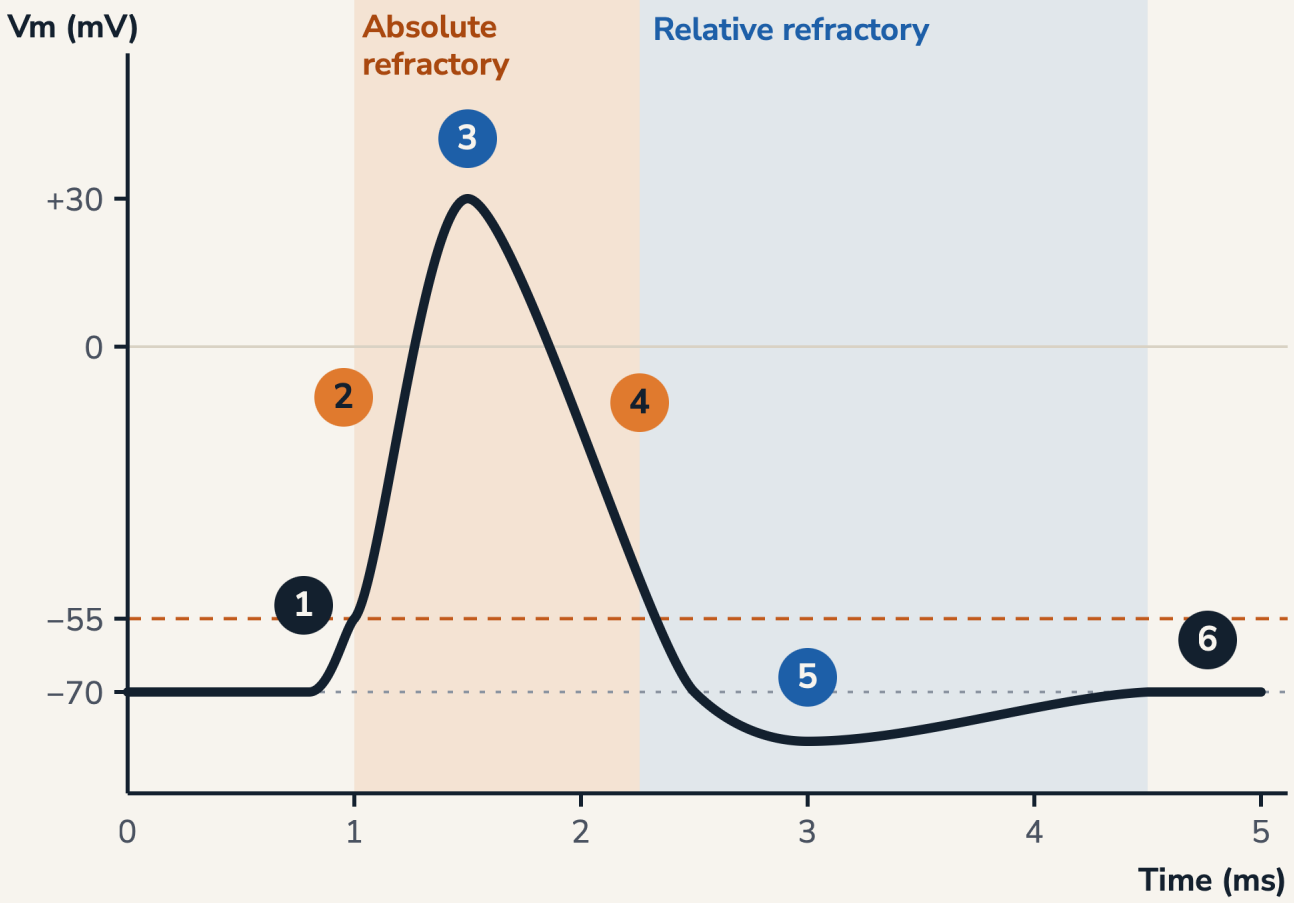
$[K^+] \approx 140$ mM inside vs 4 mM outside and $E_K \approx -90$ mV, so K^+ exits and repolarizes the cell.

PART 3

Putting it all together

Six steps, about three milliseconds, one all-or-none spike.

The sequence, step by step



1 Threshold reached
A stimulus nudges Vm from −70 to about −55 mV.

2 Na^+ channels open
Activation gates swing open; Na^+ rushes in and Vm shoots up.

3 K^+ channels open at the peak
Just before +30 mV, slow K^+ gates open as Na^+ gates inactivate.

4 Absolute refractory period
Na^+ channels are inactivated: no stimulus can fire another spike.

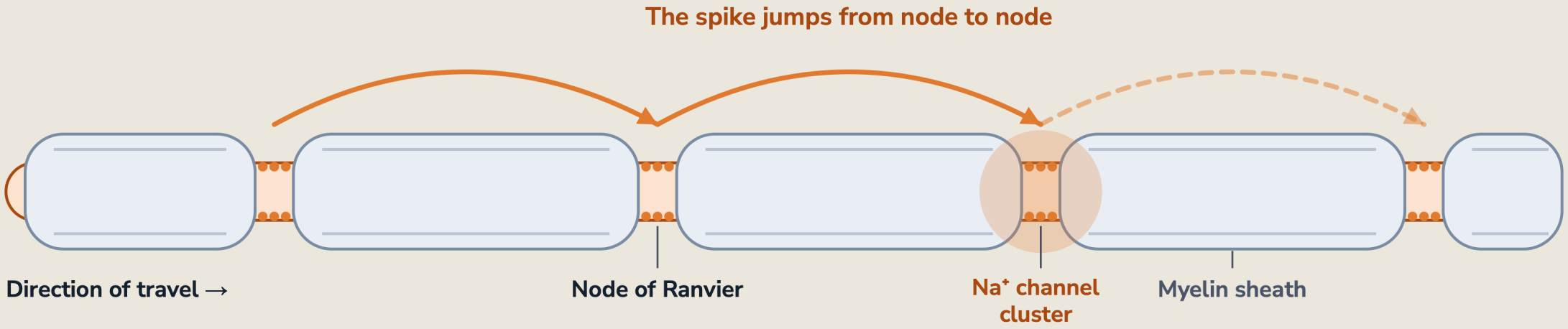
5 Relative refractory period
Na^+ gates reset but K^+ still flows: only a stronger stimulus works.

6 Back to baseline
K^+ channels close and leak channels return Vm to −70 mV.

Absolute vs relative refractory period

	Absolute refractory period	Relative refractory period
When	From threshold until most of the way through repolarization	From the end of the absolute period through the undershoot
Na^+ channels	Open, then inactivated: cannot reopen yet	Most reset to closed and ready; a few still inactivated
K^+ channels	Opening	Still open, so the membrane is hyperpolarized
Another spike?	Impossible, however strong the stimulus	Possible, with a stronger-than-normal stimulus
Why it matters	Keeps spikes separate, sets a maximum firing rate and makes travel one-way	Lets stimulus strength be coded as firing frequency

Nodes of Ranvier: saltatory conduction



Myelin insulates

Schwann cells (PNS) or oligodendrocytes (CNS) wrap the axon, cutting current leak and membrane capacitance.

Nodes regenerate the spike

Bare gaps about every 1 mm are packed with Na^+ channels, so the signal is rebuilt at full size each time.

Faster and cheaper

Up to ~120 m/s vs ~1 m/s in unmyelinated axons, and fewer ions to pump back afterward.

Four numbers to remember

−70

mV · Resting potential

Leak channels and the Na^+/K^+ pump hold it here.

−55

mV · Threshold

Na^+ channels open and the spike becomes all-or-none.

+30

mV · Peak

Na^+ gates inactivate; K^+ gates are opening.

−80

mV · Undershoot

Slow K^+ channels are still open: relative refractory.

Na^+ channels start the spike, K^+ channels end it, refractory periods keep it moving one way, and nodes of Ranvier make it fast.